

How do you find the best fit for your application?

Quality function deployment (QFD) technique is used to convert vague customer requirements into actionable parameters or specifications. However, when extended further for technical aspects, it yields many important selectors that are helpful in filtering too many available options to a very few useful ones.

In initial two to three steps of QFD, customer requirements are translated to application-level requirements. Once application requirements are understood, one can break them down further to actionable parameters for component selection. These parameters could be (but not limited to)—power consumption, operating levels, usage environment, output audio quality levels, frequency response, bandwidth, controllable frequency bands, sound clarity requirements, etc.

Typically, wherever high-quality and high-fidelity sound requirements are present, faster processors such as FPGAs are appropriate fit whilst for low-cost and basic requirements, digital or analogue processors are best fits.

—T. Anand, co-founder, Knewron

applications poses a challenge for any designer who is required to choose from so many processors and their complexities.

Let us take a look at what the audio/video processing industry has in offering for embedded multimedia designs and development of feature-rich products.

Digital and analogue solutions

Analogue processors are used where we require the best quality output. T. Anand, co-founder, Knewron explains, "The thumb rule is, where the quality of audio/video is of prime importance, we go for the analogue processing. The output from an analogue IC is much better than a digitally processed analogue output."

The latest analogue PC-controlled audio processors offer a wide range of features that are suitable for stereo and multichannel applications. These can save costs and enhance the audio signal chain of the designs. Also, these devices with integrated features such as digital volume and balance control, surround sound and tone controls, further enhance the designs.

These surround or other multichannel formats are appropriate for producing virtualised 3D sound for two-speaker systems. Also, there are 3D audio processors which can create five-speaker surround sound from a two-channel stereo source.

Although digital processing can be less, the advent of efficient and pow-

erful digital processors is an alternative to noise-prone analogue processing. Praveen Ganapathy, director, Business Development, Texas Instruments, India says, "Anything in the real world is analogue, so traditionally we could do processing in analogue domain; the only challenge is analogue domain is prone to a lot of noise. So we take the inputs in analogue form and then convert them into digital, process in the digital domain and then again convert to analogue form for the output."

Digital audio processors offer more versatile handling of audio/video streams. Echo cancellation and noise suppression DSP software technology is enhancing the audio quality of wireless products while increasing versatility in multimedia application processor families is allowing developers to design a wide range of end products with minimal incremental PCB design effort. Now you can switch sample rates without changing coefficient and have more flexible designs with the new simplified multichannel designs in digital audio processors.

Modernised system on chips (SoC) solutions

On one hand there is a wide range of fully integrated smart TV SoCs, supporting full HD through the high resolution best suited for 3D graphics and 3D gaming. On the other hand there are scalable processors with affordable ARM architecture solutions. These offer a broad range of performance, price and power consumption

to meet just about every need, and also include video accelerators, advanced graphics and display capabilities and high-speed connectivity with a rich set of peripherals that are optimised for a broad spectrum of digital video end equipment.

Consumer. SoCs with ARM architectures plus built-in hardware video accelerator engines are the solutions for applications such as tablets and mobile phones. "The mobile phone budget cannot afford so many dedicated chipsets for analogue-to-digital (A/D) conversion, codecs and for transmission. When you talk of an SoC, it develops a balance between the cost, performance and power, providing the complete ecosystem with the associated software at a good price that can fit into the mobile phone or a tablet," says Avinash Babu, senior architect, Mistral Solutions.

Automotive. For automotive multimedia designs which have been gaining popularity, the multi-tuner RF CMOS single-chip solutions with embedded AM, FM and DAB tuners offer a combined car radio and audio system fully integrated on a single IC. These can help you build the system with significantly reduced system costs via a reduced bill of material (BOM).

Time to market is also a very important factor for the designers. Gaurav Kapoor, sales manager, Intersil Corp, India, says, "There are simple low-cost controllers that help designers to kick start design and time to market, as there are no software protocols required since these devices can run by setting the simple register mapping."

These controllers can be best suited for simple automotive safety applications such as rear-view camera display where, instead of having complex software-based controllers for the interaction with the display devices, you can use these dedicated LCD controllers for straight connection with digital LCD panel. You can thus make the solution ready within a couple of weeks and production ready within a few months because of simpler designing.

Critical applications. Talking